CLAIM 2024 Checklist — Completed for This Study

Study: A Rigorous and Reproducible Benchmark of CNN and Transformer Architectures for Bone Fracture Detection on FracAtlas

Authors: Yassine Zouair, Nawal Bouknani

Reporting guideline: Checklist for Artificial Intelligence in Medical Imaging (CLAIM), 2024 Update (Tejani, Klontzas, Gatti, Mongan, Moy, Park, Kahn; *Radiology: Artificial Intelligence* 2024;6(4):e240300).

This checklist documents where each applicable CLAIM item is addressed in the manuscript. Section names refer to the submitted manuscript. Items that do not apply to a retrospective benchmark on publicly available datasets are marked **N/A***with a brief justification.*

Title and Abstract

#	Item	Status	Location / Note
1	Identification as a study of AI methodology, specifying the category of technology used (e.g., deep learning)	Yes	Title; Abstract (deep learning benchmark)
2	Structured summary of study design, methods, results, and conclusions	Yes	Abstract (Background / Methods / Results / Conclusions)

Introduction

#	Item	Status	Location / Note
3	Scientific and clinical background, including the intended use and clinical role of the AI approach	Yes	Introduction (triage and decision support for fracture detection)
4	Study objectives and hypotheses	Yes	Introduction (five stated contributions; benchmark objective)

Methods — Study Design

#	Item	Status	Location / Note
5	Prospective or retrospective study	Yes	Retrospective, using public datasets (Methods — Dataset)
6	Study goal (e.g., model creation, evaluation)	Yes	Model evaluation / benchmarking (Introduction; Methods)

Methods — Data

#	Item	Status	Location / Note
7	Data sources	Yes	FracAtlas and GRAZPEDWRI-DX, both public and cited (Methods — Dataset; External validation)
8	Inclusion and exclusion criteria	Yes	All FracAtlas images used; exact duplicates removed (Methods — Duplicate removal)
9	Data pre-processing steps	Yes	Methods — Preprocessing (resize 224 × 224, ImageNet normalization, augmentation)

#	Item	Status	Location / Note
10	Selection of data subsets, if applicable	Yes	Stratified 70/15/15 split, fixed seed (Methods — Data splitting)
11	De-identification methods	N/A	Both datasets are publicly released in fully de-identified form by their original providers; no identifiable data were handled by the authors
12	How missing data were handled	N/A	No missing image-level labels in the binary classification task; all included radiographs carry a fracture/non-fracture label
13	Definition of ground-truth reference standard	Yes	FracAtlas expert annotations; GRAZPEDWRI-DX `fracture_visible` field (Methods — Dataset; External validation)
14	Rationale for choosing the reference standard	Yes	Original expert-annotated labels from the source datasets
15	Source of ground-truth annotations; qualifications of annotators	Yes	FracAtlas annotations were performed by radiographers and verified by expert radiologists as described in Abedeen et al. (*Scientific Data* 2023); GRAZPEDWRI-DX labels were provided by pediatric radiologists per Nagy et al. (*Scientific Data* 2022)
16	Annotation tools	N/A	Labels taken as provided by source datasets; no new annotation performed
17	Measurement of inter- and intrarater variability	N/A	No new annotation performed by the authors

Methods — Data Partitions

#	Item	Status	Location / Note
18	Intended sample size and how it was determined	Yes	Full FracAtlas dataset ($n = 4{,}083$; 4,068 after duplicate removal); no subsampling
19	How data were assigned to partitions	Yes	Stratified random split, fixed seed 42 (Methods — Data splitting). Image-level partitioning; limitation discussed
20	Level at which partitions were disjoint (e.g., image, patient)	Yes	Image level, owing to absence of patient identifiers; explicitly stated and discussed in Limitations

Methods — Model

#	Item	Status	Location / Note
21	Detailed description of model	Yes	Methods — Architectures (ResNet-50, EfficientNet-B4, ViT-B/16, ConvNeXt-Tiny via `timm`)
22	Software libraries, frameworks, and packages	Yes	PyTorch / `timm` (Methods); full environment specification (`requirements.txt` / `environment.yml`) in the GitHub repository
23	Initialization of model parameters	Yes	ImageNet-pretrained weights (Methods — Architectures; Training)

Methods — Training

#	Item	Status	Location / Note
24	Details of training approach	**Yes**	Methods — Training (class-weighted BCE with $w_{pos} = n_{neg}/n_{pos}$, AdamW, cosine annealing, AMP, gradient clipping, early stopping, seed 42, single NVIDIA T4 GPU)
25	Method of selecting the final model	**Yes**	Best validation-AUC checkpoint retained (Methods — Training)
26	Details of ensembling, if used	**N/A**	No ensembling; single model per architecture

Methods — Evaluation

#	Item	Status	Location / Note
27	Metrics of model performance	**Yes**	AUC, sensitivity, specificity, PPV, NPV, F1, Expected Calibration Error (Methods — Evaluation, Calibration)
28	Statistical measures of significance and uncertainty (e.g., confidence intervals)	**Yes**	95% percentile bootstrap CIs (1,000 iterations); pairwise DeLong tests with Holm-Bonferroni correction (Methods — Evaluation)
29	Robustness or sensitivity analysis	**Yes**	Three complementary analyses: (1) stratified 5-fold cross-validation (Methods §3.7; Results §4.3); (2) leave-them-out sensitivity analysis for pHash-flagged near-duplicates showing ΔAUC = 0.009, of the same order as the CV standard deviation (Limitations §5.1); (3) zero-shot external validation on GRAZPEDWRI-DX (Methods §3.12; Results §4.6)
30	Methods for explainability or interpretability	**Yes**	Grad-CAM, with stated limitations regarding partial invariance to model reparameterization and sensitivity to target-layer choice (Methods — Interpretability; Results)
31	Validation or testing on external data	**Yes**	Zero-shot external validation on GRAZPEDWRI-DX (n = 20,327; Methods — External validation; Results)

Results

#	Item	Status	Location / Note
32	Flow of participants/cases, using a diagram to indicate inclusion/exclusion	**Yes**	Case counts reported in text (4,083 raw → 4,068 after duplicate removal → 2,848 train / 610 validation / 610 test); flow diagram provided as Supplementary Figure S1
33	Demographic and clinical characteristics of cases in each partition	**Partial**	Fracture prevalence per split reported (Methods — Data splitting); detailed demographics limited by source dataset metadata (FracAtlas does not release patient-level demographic fields)
34	Performance metrics for each data partition	**Yes**	Test-set (Table 1); cross-validation (Table 3); external cohort (Results §4.6)
35	Estimates of diagnostic accuracy and their precision	**Yes**	AUC with 95% CIs (Table 1); confusion matrices (Figure 2)

#	Item	Status	Location / Note
36	Failure analysis of incorrectly classified cases	**Yes**	Grad-CAM failure-mode analysis for false positives and false negatives (Results — Interpretability)

Discussion

#	Item	Status	Location / Note
37	Study limitations, including potential bias, statistical uncertainty, and generalizability	**Yes**	Limitations §5.1 (dataset size and class imbalance; image-level split with pHash-based near-duplicate audit and leave-out sensitivity check; single training seed; single external cohort; ViT training-recipe caveat also discussed in §5)
38	Implications for practice, including the intended clinical role	**Yes**	Discussion (triage vs. confirmatory vs. autonomous use; concrete false-positive rate at operating threshold; decision-support framing; deployment considerations including model size)

Other Information

#	Item	Status	Location / Note
39	Registration number and registry name, if applicable	**N/A**	Not a registered clinical trial (retrospective public-dataset benchmark)
40	Where the full study protocol can be accessed	**Yes**	Code and configuration: GitHub repository (`https://github.com/21yassinezouair/fracture-ai`); model checkpoint archived on Zenodo (`https://doi.org/10.5281/zenodo.21500630`); this CLAIM checklist archived in the same repository
41	Sources of funding and role of funders	**Yes**	Funding declaration (doctoral scholarship awarded to Y.Z.; no dedicated project funding; funder had no role in study design, analysis, or reporting)

Notes on Completion

- This checklist reflects the study as reported in the submitted manuscript.
- Items marked **N/A** are justified above and stem from the use of publicly available, pre-de-identified, pre-annotated datasets rather than newly collected clinical data.
- "**Partial**" indicates the item is addressed in text but not with the full formalism (e.g., detailed patient-level demographics) that a primary clinical study might include; this is appropriate for a retrospective benchmark on public data whose original metadata does not expose those fields.
- The corresponding manuscript section is indicated for each "**Yes**" item to facilitate reviewer verification.

Item numbering follows the structure of the CLAIM 2024 Update. Authors have verified item wording against the official CLAIM 2024 publication (Radiology: AI 2024;6(4):e240300).

Checklist completed by: Yassine Zouair, Corresponding Author
Affiliation: L3S Laboratory, Université Mohammed VI des Sciences de la Santé (UM6SS), Casablanca, Morocco
Email: yzouair2@um6ss.ma
ORCID: 0009-0001-5636-9187
Date: 19 August 2026
Manuscript version: v4 (final, prepared for submission to *Current Medical Imaging*)